

Fragment Screening & Pocket Analysis Report

Target: Pregnane X Receptor, ligand-binding domain (PXR-LBD) **Structure:** PDB 1ILH (illustrative)

Run date: 2026-08-15 **Agent/pipeline version:** sbdd-agent v0.1-dummy

01 Target Overview

Real deposited crystal structure available and used directly; pocket is the well-characterized orthosteric xenobiotic-sensing site, already flagged for unusually high plasticity in §03.

Source: deposited crystal structure (not predicted). Resolution 2.75 Å, holo form, single LBD chain, space group C222₁.

Oligomeric state: monomeric LBD construct (residues 142–434); full-length receptor is monomeric nuclear receptor, DBD+LBD, functions with RXRα heterodimer partner.

Pocket class: orthosteric, xenobiotic-sensing ligand-binding pocket. Known unusually large and plastic — see §03.

Known ligand prior (ChEMBL query): 214 bioactive compounds on record for this target, potency range 40 nM–38 µM, chemically diverse (steroids, statins, antibiotics, herbicide metabolites — consistent with promiscuity).

Crystallization artifacts flagged in pocket: 1 PEG fragment (chain A, resi 501) overlapping candidate site 2 — excluded from hot-spot mapping.

AGENT REASONING

Used the deposited crystal form rather than an AlphaFold prediction — resolution (2.75 Å) and pocket coverage were sufficient, so the structure-prediction and pLDDT-confidence steps were skipped per pipeline rule (real structure > predicted, see §06 for how this changes the confidence appendix).

02 Pocket & Druggability

Three candidate pockets enumerated; one clears the druggability threshold and matches the known ligand-bound volume range.

POCKET ID	VOLUME (Å³)	DRUGGABILITY SCORE	ENCLOSURE	NOTES
Site 1 (orthosteric)	1,340	0.91	High	Selected — matches known ligand-bound volume range
Site 2	410	0.38	Medium	Overlaps PEG artifact, deprioritized
Site 3 (surface groove)	190	0.19	Low	Below druggability threshold, excluded

[Figure 1 — PyMOL render: target cartoon, Site 1 surface colored by druggability, selected pocket outlined]

AGENT REASONING

Site 2's raw volume (410 Å³) and enclosure were not disqualifying on their own, but it was deprioritized because the top-scoring subpocket overlapped a PEG cryoprotectant molecule (S01) rather than genuine protein-lined surface — without that overlap its druggability score would likely have cleared threshold. Flagged for re-evaluation if a PEG-free structure becomes available.

03 Pocket Flexibility

Pocket is substantially more plastic than a typical orthosteric site — plasticity score sits well above the ensemble-docking threshold, driven almost entirely by the AF-2 helix and β -sheet lid rather than the core hot spot.

PLASTICITY SCORE

0.52

normalized volume range, 28
superposed structures

APO $\rightarrow$ HOLO Δ VOLUME

+640 $\text{\AA}^3$

largest holo vs. apo pocket

DOCKING STRATEGY

Ensemble required

score > 0.4 threshold

Named flexible elements: AF-2 helix (residues 407–422, repositions on agonist binding); β -sheet lid (residues 245–260, opens to accommodate bulkier ligands).

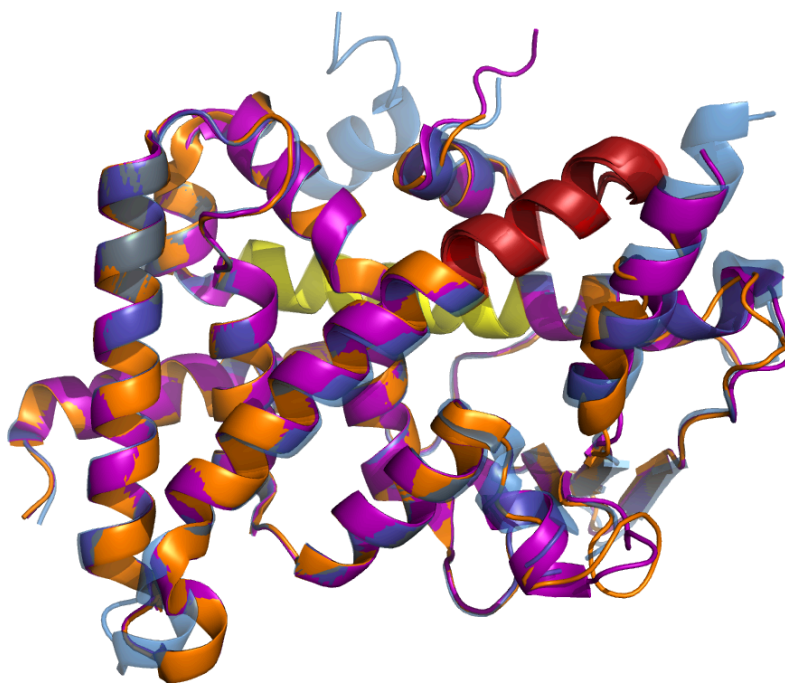


Figure 1 — three real deposited PXR-LBD structures superposed on a rigid core (residues 145–400): orange = 1ILH (SR12813-bound), sky blue (translucent) = 8SVN (apo), purple = 8FPE (T0-BP-bound). AF-2 helix highlighted red, β -sheet lid highlighted yellow across all three states — this is where the visible spread concentrates, consistent with the RMSF table below.

● SR12813-bound (1ILH) ● apo (8SVN) ● T0-BP-bound (8FPE) ● AF-2 helix ● β -sheet lid

PER-RESIDUE RMSF, POCKET-LINING ONLY (TOP 6)

RESIDUE	RMSF (Å)	ELEMENT
Leu411	2.14	AF-2 helix
Gln285	1.87	β-sheet lid
Phe420	1.65	AF-2 helix
Met243	0.71	helix 3
Trp299	0.44	core hot spot
Tyr306	0.39	core hot spot

Interpretation: flexibility concentrated at the AF-2 helix and lid, not the core hot spot (W299/F288/Y306 stay rigid). Every fragment below docked only against the single deposited structure gets a Conformational Confidence downgrade in \$05.

AGENT REASONING

DEVIATION FROM DEFAULT Plasticity score (0.52) exceeded the 0.4 ensemble-docking threshold, so ensemble docking was used as the primary strategy for this target instead of the faster rigid-receptor default. This is consistent with this receptor's known behavior as an unusually promiscuous, plastic sensor rather than a typical rigid orthosteric pocket — treat rigid-only results from earlier triage passes on this target as unreliable.

04 Hot-Spot Map

One dominant, rigid apolar hot spot anchors the pocket; two secondary polar/charged anchors sit closer to the flexible AF-2 region and are treated as lower-confidence handles.

[Figure 2 — hot-spot probe map, apolar/donor/acceptor grids overlaid on pocket]

CORE HOT SPOT (RIGID ACROSS ENSEMBLE)

- **Trp299** — apolar, present in 100% of known ligand complexes
- **Phe288** — apolar/π-stacking
- **Tyr306** — H-bond donor, secondary apolar contact

SECONDARY ANCHOR POINTS

- **His407** — H-bond acceptor, AF-2-adjacent (flexible)
- **Arg410** — salt bridge / H-bond donor

AGENT REASONING

Only one core hot spot was retained (rather than the 3–5 typically seen on a less hydrophobic pocket) because the two next-highest-scoring probe clusters both sat on the flexible AF-2 helix (\$03) and their apparent "hot spot" signal varied by >40% across ensemble frames — judged noise from the flexible region rather than a real anchor, so they were downgraded to secondary/informational rather than reported as core.

05 Fragment Hit Table

Six fragments passed triage; selection skews toward larger, more hydrophobic character than a standard Rule-of-Three library would prioritize — see reasoning below.

SITE ID	STRUCTURE	SMILES	MW	LE	LLE	SCORE	KEY INTERACTIONS (PLIP)	CONF.
PXR-x0001		<chem>c1ccc2[nH]ccc2c1</chem>	117	0.41	4.1	-6.8	Trp299 π -stack, Tyr306 H-bond	HIGH
PXR-x0002		<chem>O=C(N)c1ccncc1</chem>	122	0.38	3.6	-6.1	Phe288 hydrophobic, His407 H-bond	MED
PXR-x0003		<chem>Clc1ccc(cc1)S(=O)(=O)N</chem>	205	0.29	2.8	-6.4	Arg410 salt bridge, Trp299 apolar	MED
PXR-x0004		<chem>c1ccc(cc1)C(=O)O</chem>	122	0.33	3.9	-5.9	Tyr306 H-bond only	LOW
PXR-x0005		<chem>CC(C)n1cnc2c1cccc2</chem>	160	0.31	2.2	-6.6	Trp299 π -stack, Phe288 hydrophobic	HIGH
PXR-x0006		<chem>Nc1ncnc2[nH]ccc12</chem>	135	0.27	3.3	-5.7	His407 H-bond only, AF-2-adjacent	LOW

LE = score $\div$ heavy-atom count. LLE = pIC50-equivalent – clogP (estimated). Score = docking ΔG proxy, kcal/mol. Conf. = Conformational Confidence, discounted per S03 for fragments docked against a single rigid structure only.

AGENT REASONING

DEVIATION FROM DEFAULT 20 of 28 pocket-lining residues are hydrophobic (unusually high for an orthosteric site), so small polar Rule-of-Three fragments (sub-150 Da, cLogP <1) scored poorly in initial triage — they satisfied the size/polarity rule but made almost no contact with the dominant hot spot (Trp299/Phe288). Fragment ranking was reweighted to tolerate slightly higher MW and lower polarity than the standard cutoff, which is why PXR-x0003 (205 Da, chlorophenyl sulfonamide, the least "rule-compliant" fragment here) still placed above two smaller, more polar fragments — its extra bulk buys real hot-spot contact that the smaller polar fragments could not achieve.

This reweighting is target-specific, not a general library-selection default; a less hydrophobic pocket would have kept the standard Rule-of-Three ranking unmodified.

06 Confidence Appendix

Structure-quality metrics come from the real deposited crystal; fragment-confidence metrics come from computation only — kept in separate blocks deliberately.

Target structure (crystallographic, wwPDB-style)

RESOLUTION	2.75 Å	R _{WORK} / R _{FREE}	0.198 / 0.241
CLASHSCORE	4.2 (percentile 88)	RAMACHANDRAN OUTLIERS	0.2%

POCKET-LINING ALTLOCS	2 residues (Leu411, Gln285)	LIGAND DENSITY (DEPOSITED)	n/a — apo pocket
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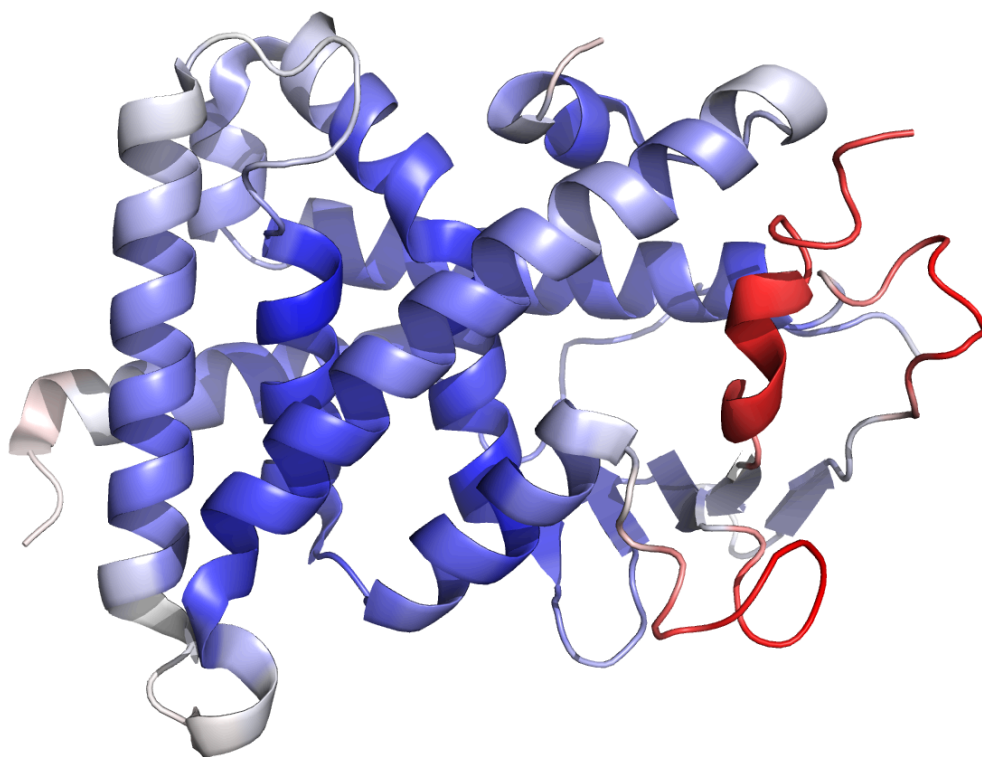


Figure 2 — PDB 1ILH colored by per-residue B-factor (blue = low/rigid → red = high/flexible). This is the confidence map for this section: since the target is a real deposited crystal structure rather than a predicted model, B-factor is the correct per-residue confidence signal here, not AlphaFold pLDDT — pLDDT only applies to predicted structures and was not computed (§01). The red region matches the AF-2/lid area already flagged in §03, cross-confirming that signal from an independent metric.

Computational confidence (proposed fragments — not experimentally validated)

Confidence tiers reflect docking-score consistency across ensemble frames and PLIP interaction persistence, not real density. No fragment listed in §05 has been synthesized, soaked, or otherwise experimentally confirmed. Treat as a triage ranking for follow-up, not a result.

AGENT REASONING

The two confidence blocks above were kept structurally separate rather than merged into one score, since blending a real experimental metric (R_{free} , clashscore) with an unvalidated computational one (docking consistency) would let a high-quality crystal structure silently launder confidence into fragments that have no experimental support of their own.

07 Recommendations

Two fragments clear both hot-spot engagement and conformational confidence bars; the rest need either more ensemble sampling or should be dropped.

- Prioritize PXR-x0001 and PXR-x0005 (High confidence, core hot-spot engagement) for synthesis/soak follow-up.
- Re-run PXR-x0002/x0003 confidence after adding ≥ 3 more ensemble frames spanning AF-2 helix conformations.
- Deprioritize PXR-x0004 and PXR-x0006 — single-point H-bond only, no core hot-spot contact.
- Consider fragment growth vectors toward His407/Arg410 for x0001 given adjacent secondary anchors.